

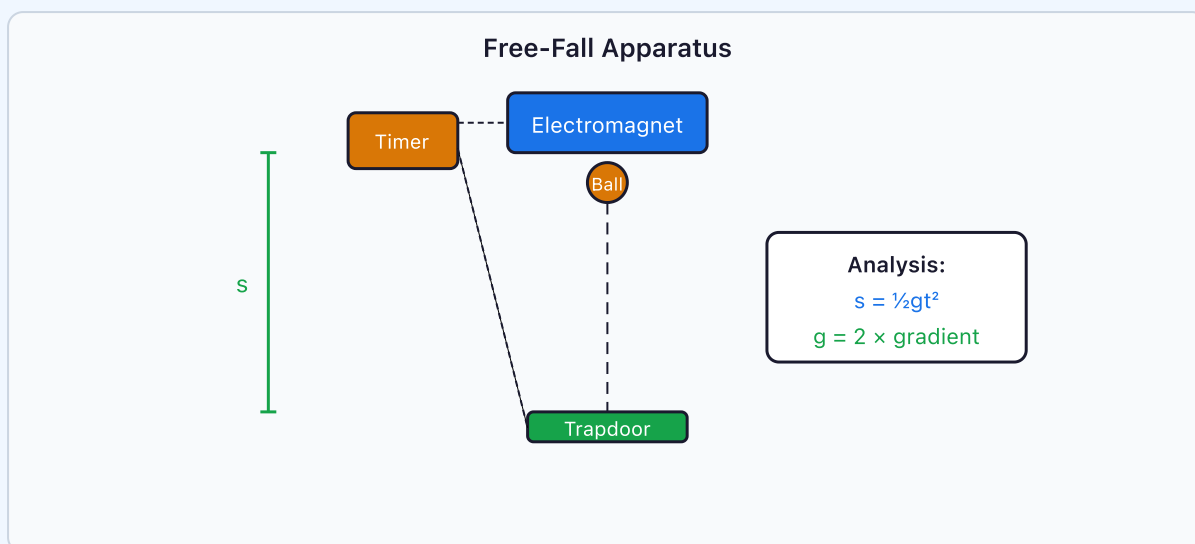
OCR A A-Level Physics — Practical Activity Groups (PAGs)

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OCR H556 requires at least 12 practical activities covering PAG1–PAG12 for the Practical Endorsement. Each PAG is a group of techniques; the experiment shown is a typical activity, and centres may use others.

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PAG1 – Investigating Motion: Determination of g by Free Fall



Apparatus

- Electromagnet with release switch
- Steel ball bearing
- Trapdoor switch and electronic timer
- 2 m ruler, clamp stand

Method

1. Clamp electromagnet at height s above the trapdoor.
2. Release ball; timer starts when ball releases and stops when trapdoor opens.
3. Record t . Repeat 3 times at each height; take mean.
4. Vary s over at least 6 values (0.2–1.0 m).
5. Plot s vs t^2 . Gradient = $g/2$.

Safety

- Soft pad beneath trapdoor
- Stable clamp stand; feet clear of falling ball

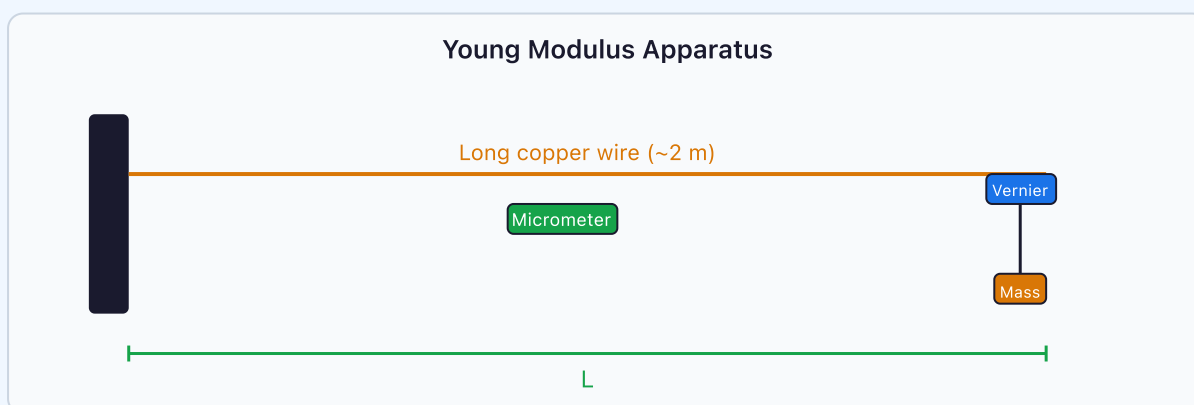
Analysis

$$s = \frac{1}{2}gt^2 \Rightarrow \text{gradient of } s \text{ vs } t^2 = g/2$$

Source of Error	Type	Improvement
Residual magnetism delays release	Systematic	Use demagnetising coil or mechanical release
Air resistance	Systematic	Use small, dense ball
Height measurement	Random	Measure from bottom of ball to trapdoor centre
Timer resolution	Random	Use high-resolution electronic timer

Exam Tip: The graph should pass through the origin. A negative y-intercept (line crossing the t^2 axis to the right of the origin) suggests a timing delay (systematic error). Calculate percentage difference from accepted $g = 9.81 \text{ m s}^{-2}$ to evaluate accuracy.

PAG2 – Investigating Properties of Materials: Young Modulus of a Metal Wire



Apparatus

- Long thin wire (~2 m copper), G-clamp
- Micrometer screw gauge, vernier scale/travelling microscope

- Slotted masses, tape measure, safety goggles

Method

1. Measure diameter at 6+ points with micrometer. Calculate $A = \pi d^2/4$.
2. Measure original length L .
3. Add masses in steps. Record F and extension ΔL from vernier scale.
4. Unload to check elastic behaviour.
5. Plot stress vs strain. Gradient = E (Young modulus).

Safety

- Safety goggles; cushion beneath masses; do not exceed elastic limit

Analysis

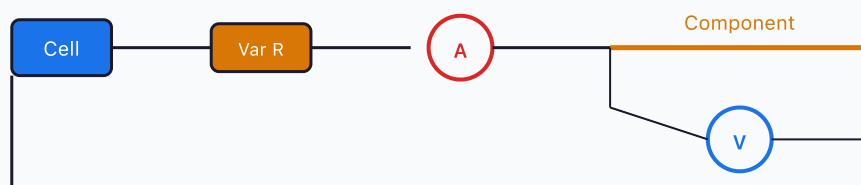
$$E = \sigma/\epsilon = (F/A) / (\Delta L/L)$$

Source of Error	Type	Improvement
Diameter varies	Random	Multiple readings at different positions
Wire kinks	Systematic	Apply small initial load
Thermal expansion	Systematic	Use reference wire
Small extensions	Random	Use long, thin wire; travelling microscope

Exam Tip: The area under the stress-strain graph up to the elastic limit represents the elastic strain energy per unit volume. The gradient of the linear region gives E . Units: Pa or N m^{-2} .

PAG3 – Investigating Electrical Properties: Resistivity of a Metal & I–V Characteristics

Resistivity & V-I Characteristics



Resistivity: R vs $L \rightarrow \rho = \text{gradient} \times A$

V-I: swap component (wire, filament, diode)

Apparatus

- **Resistivity:** Constantan wire, ammeter, voltmeter, micrometer, ruler, clips
- **V-I characteristics:** Filament lamp, diode, or resistor; variable resistor or potential divider; ammeter; voltmeter

Method (Resistivity)

1. Measure wire diameter with micrometer (6+ readings). Calculate $A = \pi d^2/4$.
2. Connect circuit with ammeter in series, voltmeter in parallel across the wire.
3. Vary length L using crocodile clips. Record V and I ; calculate $R = V/I$.
4. Plot R vs L . Gradient $= \rho/A$.

Method (V-I Characteristics)

1. Replace wire with the component of interest (filament lamp, diode, or fixed resistor).
2. Use a variable resistor (or potential divider for full range including 0 V).
3. Record V and I for a range of values. Reverse connections for diode.
4. Plot V - I graph.

Safety

- Switch off between readings to prevent overheating
- Filament lamp gets hot — do not touch
- Low voltage (< 6 V)

Analysis

Resistivity: $\rho = \text{gradient} \times A$ from R vs L graph.

V-I graphs:

- Ohmic conductor (wire): straight line through origin
- Filament lamp: curve showing increasing gradient (resistance increases with temperature)
- Diode: negligible current in reverse; rapid increase above threshold voltage (~ 0.6 V for silicon)

Source of Error	Type	Improvement
Wire heating	Systematic	Low current; switch off between readings
Contact resistance	Systematic	Clean contacts; firm connections
Parallax on analogue meters	Random	Read at eye level; use digital meters
Diameter variation	Random	Multiple readings at different points

Exam Tip: For V-I characteristics, use a potential divider circuit rather than a simple variable resistor if you need the full voltage range from 0 V upwards. The variable resistor method cannot achieve 0 V across the component.

🔧 PAG3 – Investigating Electrical Properties: EMF and Internal Resistance of a Cell

EMF and Internal Resistance



$$V = \epsilon - Ir \Rightarrow y\text{-int} = \epsilon, \text{ grad} = -r$$

Apparatus

- Cell, variable resistor, ammeter, voltmeter, switch

Method

1. Connect cell, ammeter, variable resistor in series. Voltmeter across cell.
2. Record V and I for 6+ resistance settings. Switch off between readings.
3. Plot V vs I . y -intercept = ε , gradient = $-r$.

Safety

- Do not short-circuit cell; switch off between readings

Analysis

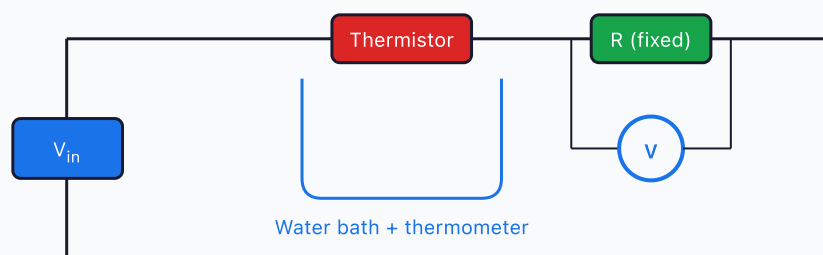
$$V = \varepsilon - Ir$$

Source of Error	Type	Improvement
Cell runs down	Systematic	Switch off between readings; work quickly
Meter resistances	Systematic	Digital meters with appropriate impedance
Fluctuating readings	Random	Wait for stability; take repeats
Temperature variation	Systematic	Stable environment

Exam Tip: Never confuse EMF (measured at open circuit, $I = 0$) with terminal p.d. (measured under load). The "lost volts" = Ir . You can find ε by extrapolating the V vs I line to $I = 0$.

PAG4 – Investigating Electrical Circuits: Potential Divider Sensing Circuits

Potential Divider with a Thermistor



$$V_{out} = V_{in} \times R / (R + R_T)$$

Apparatus

- Low-voltage DC supply (about 5–6 V) or battery
- NTC thermistor and/or LDR; fixed resistor (or variable resistor) of similar resistance
- Digital voltmeter (multimeter), connecting leads, crocodile clips
- **Thermistor:** beaker of water, kettle or heater, ice, thermometer, stirrer
- **LDR:** lamp, metre ruler, darkened room (or a light meter)

Method

1. Connect the thermistor and fixed resistor in series across the supply. Measure the supply p.d. V_{in} .
2. Connect the voltmeter across the fixed resistor to measure the output p.d. V_{out} .
3. Put the thermistor (leads kept dry, e.g. in a thin sealed sleeve) in iced water next to the thermometer.
4. Warm the water in steps of about 10 °C. Stir, wait until V_{out} is steady, then record θ and V_{out} .
5. Repeat the readings as the water cools and take means.
6. For the LDR, change the light level by moving the lamp to a range of distances in a darkened room and record V_{out} at each.

Safety

- Hot water can scald — stand the beaker on a heat-proof mat and do not move it while hot
- Keep connections and the supply away from water
- Low voltage only; switch off between readings to limit self-heating

Analysis

$$V_{\text{out}} = V_{\text{in}} \times R / (R + R_T) \Rightarrow R_T = R(V_{\text{in}} - V_{\text{out}}) / V_{\text{out}}$$

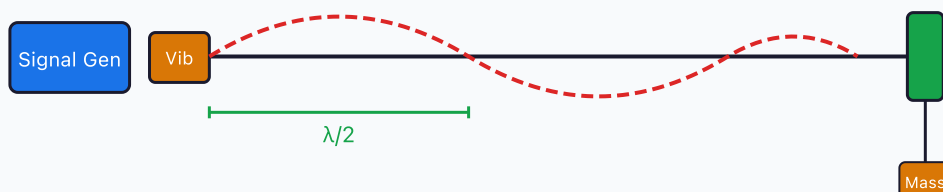
For an NTC thermistor, R_T falls as temperature rises, so the p.d. across the fixed resistor rises. Plot V_{out} against θ (or R_T against θ); the curve is non-linear, and the circuit is most sensitive where R_T is close to R .

Source of Error	Type	Improvement
Thermistor not at the water temperature	Systematic	Stir; wait for a steady reading; place thermometer next to thermistor
Self-heating of thermistor by the current	Systematic	Low supply p.d.; larger series resistor; brief readings
Supply p.d. changes (internal resistance)	Systematic	Measure V_{in} at every reading
Reading the thermometer	Random	Read at eye level or use a digital thermometer; repeat on cooling

Exam Tip: Swapping the sensor and the fixed resistor reverses the behaviour (the output then falls as temperature rises). A voltmeter or next stage connected to the output must have a resistance much larger than R , otherwise it loads the divider and lowers V_{out} .

PAG5 – Investigating Waves: Stationary Waves on a String

Stationary Waves on a String



Other PAG5 activities in the spec: OCR's own example is finding the wavelength of light with a double slit and a diffraction grating (4.4.3). The spec also tags the speed of sound from stationary waves in a resonance tube (4.4.4), using an oscilloscope to find frequency, and polarisation with microwaves and light as PAG5.

Apparatus

- Signal generator and vibrator
- String, pulley, bench clamp
- Slotted masses, metre ruler

Method

1. Attach string to vibrator, pass over pulley with hanging mass for tension.
2. Adjust frequency for fundamental mode. Record f and measure L .
3. Find higher harmonics. Record frequency and number of loops.
4. Vary tension or string type and repeat.

Safety

- Safety screen; soft pad beneath masses

Analysis

$$f = (1/2L)\sqrt{(T/\mu)}$$

Plot f vs $1/L$ at constant T : straight line through origin.

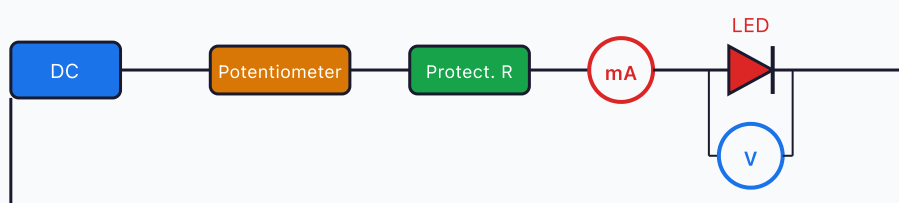
Source of Error	Type	Improvement
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Judging resonance by eye	Random	Use paper rider on string
L measurement	Random	Measure between nodes, not from vibrator
Mass hanger mass omitted	Systematic	Include hanger in tension calculation
String not uniform	Random	Weigh a measured length to find μ accurately

Exam Tip: At resonance, the amplitude is maximum and the pattern is stable. The number of antinodes equals the harmonic number. Wavelength = $2L/n$ for the n th harmonic.

🔧 PAG6 – Investigating Quantum Effects: Planck Constant Using LEDs

LED Circuit for the Planck Constant



$$eV = hc/\lambda \Rightarrow \text{plot } V \text{ vs } 1/\lambda, \text{ gradient} = hc/e$$

Apparatus

- Set of LEDs of different colours, each with a known (manufacturer's) peak wavelength
- DC supply with potentiometer (or variable resistor), protective series resistor (a few hundred ohms)
- Digital voltmeter and milliammeter
- Short black tube to view the LED; darkened room

Method

1. Connect the LED in forward bias in series with the protective resistor and milliammeter; connect the voltmeter directly across the LED.
2. Starting from 0 V, slowly increase the p.d. until the LED just begins to glow, viewing it through the black tube. Record this threshold p.d. V.
3. Repeat three times for each LED and find the mean V.
4. Alternatively, record current against p.d. and extrapolate the straight, conducting part of the I–V graph back to $I = 0$ to find V.
5. Repeat for every LED and tabulate λ and $1/\lambda$.

Safety

- Always include the protective resistor so the LED's maximum current is not exceeded
- Do not stare into bright LEDs, especially blue or violet ones
- Low voltage

Analysis

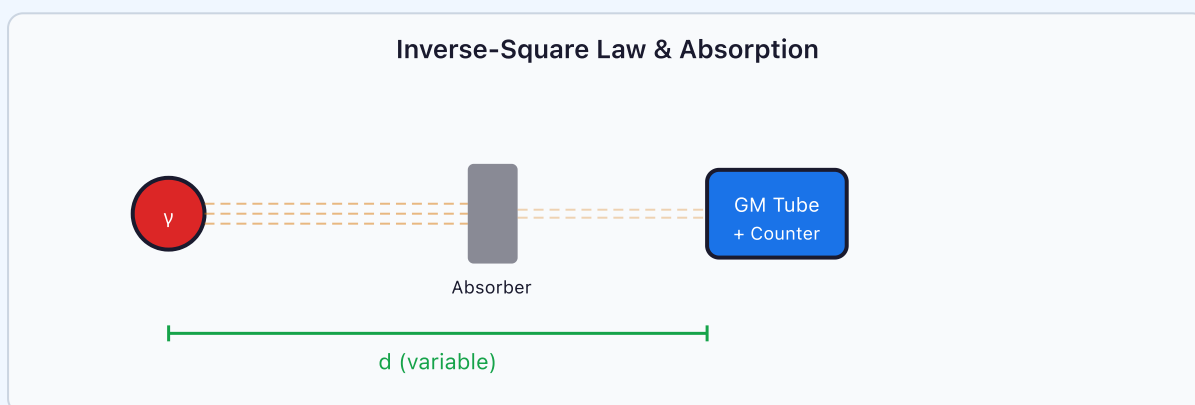
$$eV = hc/\lambda \Rightarrow V = (hc/e) \times (1/\lambda)$$

Plot V against $1/\lambda$. Gradient = hc/e , so $h = \text{gradient} \times e / c$. Compare with the accepted value $h = 6.63 \times 10^{-34} \text{ J s}$.

Source of Error	Type	Improvement
Judging when the LED first glows	Random	Darken the room; view through a black tube; use the I–V extrapolation method
LED emits a band of wavelengths, not one	Systematic	Use the manufacturer's peak wavelength; choose narrow-band LEDs
Some energy lost inside the LED	Systematic	Use the gradient, not single points, so a constant offset does not matter
Voltmeter resolution	Random	Digital voltmeter reading to 0.01 V; repeat readings

Exam Tip: Each photon emitted is produced by one electron transferring energy eV, so $eV = hf = hc/\lambda$. The graph of V against $1/\lambda$ may not pass through the origin; this is why h is found from the gradient rather than from a single LED.

PAG7 – Investigating Ionising Radiation: Inverse-Square Law & Absorption of Gamma Radiation



Other PAG7 activities in the spec: OCR's own example is the absorption of α , β or γ radiation (6.4.3(b)(ii)), and the spec also tags finding the half-life of protactinium (6.4.3(e)(ii)) as PAG7. The inverse-square law for γ -rays is a common PAG7 activity but is not itself a spec learning outcome.

Apparatus

- Gamma source (sealed), GM tube and counter
- Metre ruler, lead absorbers of various thicknesses
- Source holder, clamp

Method (Inverse-Square Law)

1. Measure background count rate (60+ seconds, no source).
2. Place source at distance d from GM tube. Record count rate over 60 s.
3. Repeat for 6+ distances. Take 3 readings at each.

4. Subtract background. Plot corrected count rate vs $1/d^2$.

Method (Absorption)

1. Fix source-detector distance. Measure count rate with no absorber.
2. Insert lead sheets of known thickness. Record count rate for each thickness.
3. Plot $\ln(\text{corrected count rate})$ vs thickness. Gradient = $-\mu$.

Safety

- Tongs for source; maximise distance; minimise time
- Point source away from people; lead-lined storage
- ALARA principle

Analysis

Inverse-square: $I \propto 1/d^2$ — C vs $1/d^2$ should be linear through origin.

Absorption: $I = I_0 e^{-\mu x}$ — $\ln(C)$ vs x gives gradient = $-\mu$.

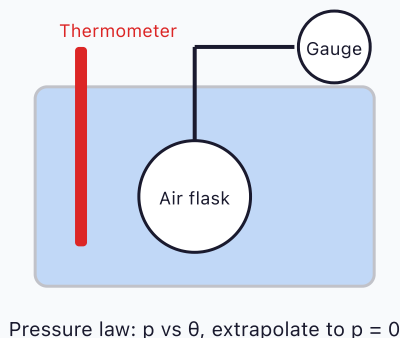
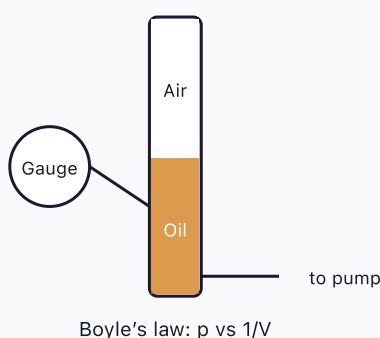
Half-value thickness: $x_{1/2} = \ln 2 / \mu$.

Source of Error	Type	Improvement
Random nature of decay	Random	Count for longer; take repeats
Background fluctuations	Random	Long background measurement; multiple readings
Distance to wrong part of tube	Systematic	Measure to GM tube window centre
Absorber thickness not uniform	Systematic	Measure thickness at multiple points

Exam Tip: Always subtract background before any calculations. For the inverse-square law, the source must approximate a point source, which is valid when d is much larger than the source diameter. Plotting $\ln(C)$ vs $\ln(d)$ should give a gradient of -2 .

PAG8 – Investigating Gases: Boyle's Law and an Estimate of Absolute Zero

Gas Law Investigations



Apparatus

- **Boyle's law:** sealed glass tube with a column of air trapped above oil, pressure gauge, pump with valve, scale alongside the tube
- **Pressure law:** round-bottomed flask with bung and short tube to a pressure gauge (or pressure sensor and data logger)
- Large beaker or water bath, heater, ice, thermometer, stirrer
- Eye protection and safety screen

Method (Boyle's Law, constant temperature)

1. Use the pump to push oil up the tube and compress the trapped air; close the valve.
2. Wait about a minute so the air returns to room temperature, then record pressure p and length l of the air column ($V \propto l$ for a uniform tube).

3. Repeat for at least six pressures, then release the pressure in steps and repeat.
4. Plot p against $1/V$ (or $1/l$).

Method (Pressure Law, constant volume)

1. Fully submerge the flask in iced water, keeping the tube to the gauge as short as possible.
2. Heat the bath in steps of about $10\text{ }^{\circ}\text{C}$ up to near boiling. At each step, stir, wait until the pressure is steady, and record θ and p .
3. Plot p against θ in $^{\circ}\text{C}$ and extrapolate the line back to $p = 0$.

Safety

- Glass under pressure can shatter — eye protection, safety screen, do not exceed the apparatus's rated pressure
- Hot water can scald; stable, heat-proof base

Analysis

$$pV = nRT$$

Boyle's law: at constant T and n , $pV = \text{constant}$, so p against $1/V$ is a straight line through the origin.

Pressure law: at constant V and n , $p/T = \text{constant}$. With θ in $^{\circ}\text{C}$, $p = k(\theta + 273)$, so the intercept on the θ axis (where $p = 0$) estimates absolute zero, about $-273\text{ }^{\circ}\text{C}$.

Source of Error	Type	Improvement
Air in tubing outside the bath at a different temperature	Systematic	Keep connecting tube short and narrow; submerge flask fully
Gas temperature lags behind the bath	Systematic	Stir; wait for a steady pressure reading
Compression warms the gas (Boyle's law)	Systematic	Wait after each change before reading
Reading gauge and scale	Random	Read at eye level; use a pressure sensor; repeat on the return sweep

Gas leaks

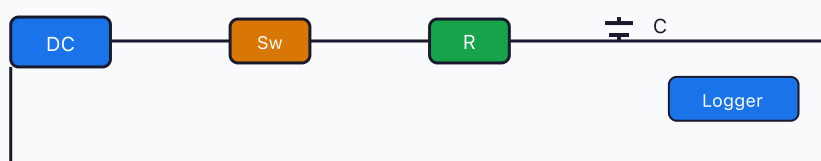
Systematic

Check seals; readings should repeat on the return sweep

Exam Tip: Always use T in kelvin and p in Pa in $pV = nRT$. On a p - θ graph in $^{\circ}\text{C}$ the line does not pass through the origin. Extrapolating a long way from the data magnifies the effect of any uncertainty in the gradient, so the estimate of absolute zero has a large uncertainty.

PAG9 – Investigating Capacitors: Charge and Discharge (Time Constant)

RC Circuit

 $\ln(V) \text{ vs } t \rightarrow \text{gradient} = -1/RC$

Apparatus

- Capacitor (470–2200 μF), resistor, DC supply
- Two-way switch, datalogger or voltmeter + stopwatch

Method

1. Connect RC circuit. Datalogger across C .
2. Charge by connecting supply. Record V vs t .
3. Switch the capacitor so it discharges through R (supply disconnected). Record V vs t .
4. Plot $\ln(V)$ vs t for discharge. Gradient = $-1/RC$.

Safety

- Correct polarity; discharge before handling; low voltage

Analysis

$$V = V_0 e^{-t/RC} \Rightarrow \ln V = \ln V_0 - t/RC$$

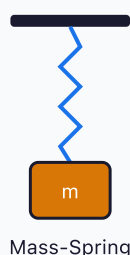
$\tau = RC$ is the time for V to fall to V_0/e (37%).

Source of Error	Type	Improvement
Voltmeter draws current	Systematic	High-impedance meter or datalogger
Manual timing	Random	Use datalogger
Capacitor leakage	Systematic	High-quality capacitor
Temperature drift	Systematic	Stable environment

Exam Tip: If the $\ln(V)$ vs t graph is not straight, this suggests the capacitance or resistance is changing during discharge (e.g. due to temperature effects). Choose R and C for $\tau = 10\text{--}30$ s.

PAG10 – Investigating Simple Harmonic Motion: Mass–Spring and Pendulum

SHM Investigations



Mass-Spring



Pendulum

Spring: T^2 vs m | Pendulum: T^2 vs L

Apparatus

- Spring, slotted masses, clamp stand, stopwatch
- String, bob, protractor, ruler

Method (Mass-Spring)

1. Hang mass m from spring. Pull down slightly, release.
2. Time 10 oscillations. Calculate T . Plot T^2 vs m .
3. Gradient = $4\pi^2/k$.

Method (Pendulum)

1. Measure L from pivot to centre of bob. Displace $< 10^\circ$, release.
2. Time 10 oscillations. Calculate T . Plot T^2 vs L .
3. Gradient = $4\pi^2/g$, so $g = 4\pi^2/\text{gradient}$.

Safety

- Stable clamp; clear of swinging masses; do not overstretch spring

Analysis

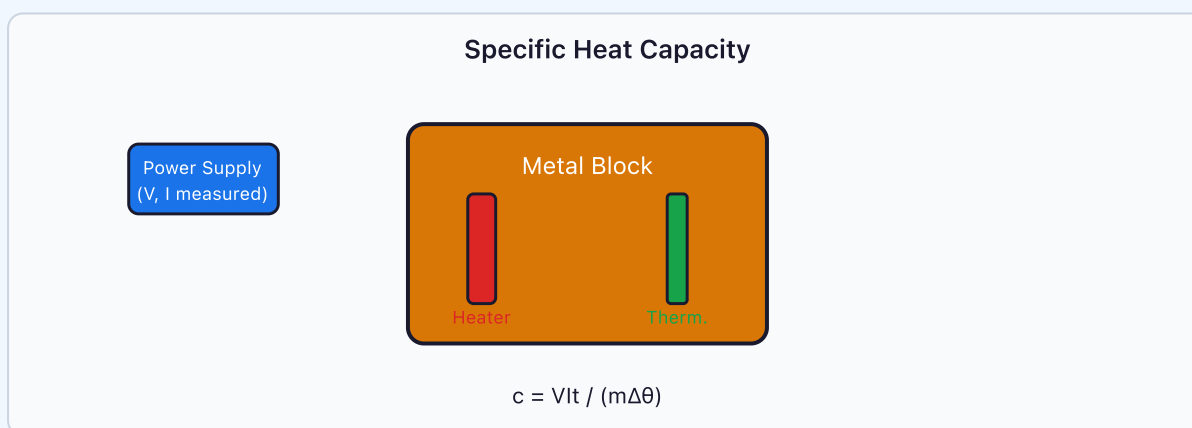
$$\text{Spring: } T^2 = (4\pi^2/k)m \quad \text{Pendulum: } T^2 = (4\pi^2/g)L$$

Source of Error	Type	Improvement
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Miscounting oscillations	Random	Use fiducial marker; count from equilibrium
Large amplitude (pendulum)	Systematic	Keep $\theta < 10^\circ$
Non-vertical oscillation (spring)	Random	Release carefully from rest
Reaction time	Random	Time 10+ oscillations

Exam Tip: The pendulum period is independent of mass and (for small angles) of amplitude. The spring period is independent of g , so it would be the same on the Moon. Always plot T^2 (not T) to linearise the graph.

PAG11 – Investigation: Specific Heat Capacity of a Metal



Apparatus

- Metal block (aluminium/copper) with holes for heater and thermometer
- Immersion heater, thermometer, ammeter, voltmeter
- Power supply, stopwatch, balance, insulation

Method

1. Weigh the block. Insert heater and thermometer (with oil for thermal contact).
2. Record initial temperature. Switch on heater.
3. Record V , I . Heat for a measured time t .
4. Record final temperature. Calculate $c = VIt / (m\Delta\theta)$.

Safety

- Hot block — use tongs; heater fully inserted; heat-resistant mat

Analysis

$$c = VIt / (m\Delta\theta)$$

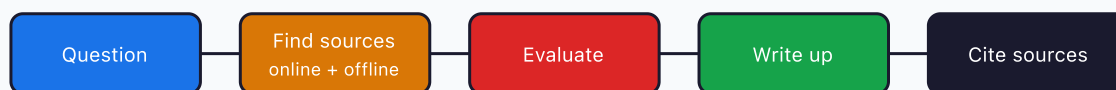
Or plot $\Delta\theta$ vs t : gradient = $VI/(mc)$.

Source of Error	Type	Improvement
Heat loss to surroundings	Systematic (c too high)	Insulate block; cooling correction
Poor thermal contact	Systematic	Use oil in thermometer hole
V/I fluctuations	Random	Monitor throughout; use regulated supply
Thermometer lag	Systematic	Wait for equilibrium

Exam Tip: Measured c is typically higher than the accepted value because of heat losses. A continuous heating method (plotting $\Delta\theta$ vs t) with a cooling correction gives better results than a single measurement.

PAG12 – Research Skills

The Research Process



Every claim you use should be traceable to a reference

Overview

PAG12 is not a single experiment. It assesses whether you can research a physics topic using both online and offline sources and correctly cite the information you use. The research can stand alone or support a practical, for example finding a suitable method before an investigation (PAG11).

Example Topics

- **How GPS works:** satellites broadcast precise time signals; a receiver finds its distance from each satellite from the signal travel time ($\text{distance} = c \times t$) and needs at least four satellites to fix position and correct its own clock error.
- **Radioactive tracers in medical imaging:** e.g. technetium-99m (a gamma emitter with a half-life of about 6 hours) for gamma-camera imaging, and fluorine-18 (a positron emitter) for PET scanning.

Method

1. Write a clear, focused research question.
2. Search a range of sources: textbooks, scientific journals and magazines, and reputable websites (universities, research organisations, government bodies).
3. Evaluate each source for reliability and relevance before using it.
4. Make notes in your own words, recording full reference details as you go.
5. Write up your findings, citing sources in the text, and finish with a reference list.

Evaluating Sources

Check	What to look for
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Authority	Who wrote it? Are they an expert or a recognised organisation?
Evidence	Is it peer-reviewed or supported by data and references?
Currency	Is it up to date for the topic?
Bias	Is the author selling something or arguing one side only?
Agreement	Do independent sources agree with it?

Referencing

Use one consistent style (for example Harvard). For each source record the author, date, title, publisher or website name, and, for web pages, the full URL and the date you accessed it. Put a short citation next to each piece of information you use, e.g. (Author, year), matching an entry in the reference list.

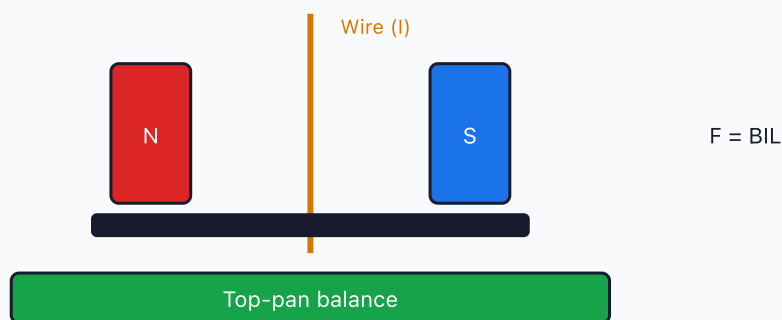
Exam Tip: Copying text without citing it is plagiarism. Paraphrase in your own words and cite the source; use quotation marks for any exact wording. Using more than one independent source lets you cross-check facts.

Supporting Practicals — Not PAG Activities for H556

The sections below are useful revision but are **not** one of OCR's twelve Practical Activity Groups. The force-on-a-wire experiment is in the specification content (6.3.1) and can appear in exams; the photocell experiment goes beyond the spec, which only expects a simple demonstration such as a gold-leaf electroscope and zinc plate (4.5.2); the data analysis skills apply to every PAG.

Force on a Current-Carrying Wire (spec 6.3.1 — not a PAG)

Force on Current-Carrying Wire



Apparatus

- Horseshoe/U-shaped magnet on iron yoke
- Copper wire "cradle", top-pan balance (0.01 g)
- DC power supply, ammeter, ruler

Method

1. Place magnet on balance. Zero balance.
2. Position wire in the magnetic field.
3. Vary current. Record I and mass change Δm .
4. $F = \Delta m \times g$. Measure wire length L in the field.
5. Plot F vs I . Gradient = BL , so $B = \text{gradient}/L$.

Safety

- Low current; short bursts; keep magnets away from electronics

Analysis

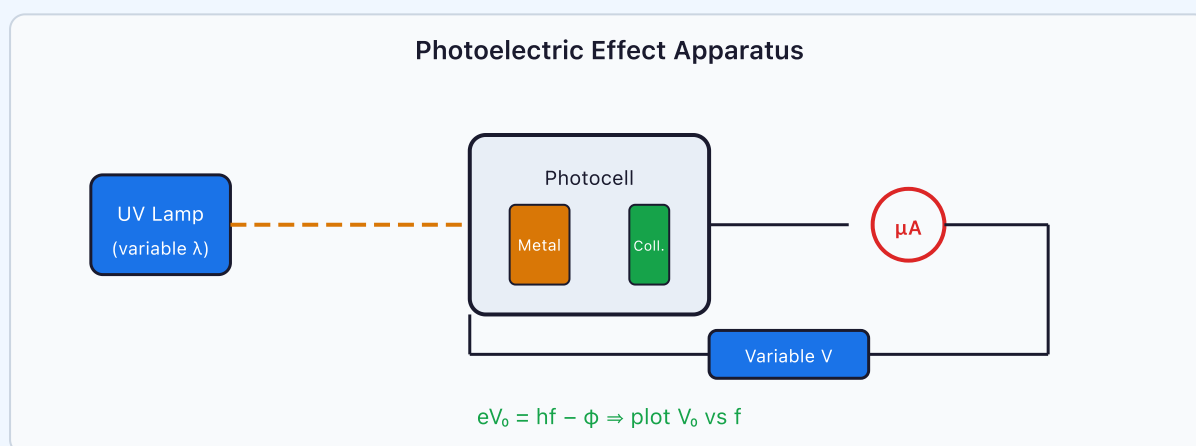
$$F = BIL \Rightarrow \text{plot } F \text{ vs } I, \text{ gradient} = BL$$

Source of Error	Type	Improvement
Draughts/vibrations affect balance	Random	Draught shield; stable surface
Non-uniform field	Systematic	Use strong flat-faced magnets

Wire not perpendicular to field	Systematic	Align carefully
Wire heating	Systematic	Short bursts; cooling between readings

Exam Tip: By Newton's third law, the force on the wire equals the force on the magnet. This is why the balance reading changes. $F = BIL$ only applies when wire is perpendicular to B . Use Fleming's left-hand rule for direction.

🔧 Photoelectric Effect with a Photocell (extension — not a PAG)



Apparatus

- Photocell (vacuum photodiode)
- UV lamp or set of LEDs with known wavelengths / colour filters
- Variable DC power supply (to apply reverse voltage)
- Microammeter (sensitive ammeter)
- Voltmeter

Method

1. Illuminate the metal cathode of the photocell with light of known frequency f .
2. Observe the photocurrent on the microammeter.

3. Gradually increase the reverse (stopping) voltage until the photocurrent drops to zero.
4. Record the stopping voltage V_0 for this frequency.
5. Repeat for at least 5 different frequencies (using filters or different LEDs).
6. Plot V_0 vs f .

Safety

- Do not look directly at UV light — use UV-blocking goggles
- Keep UV exposure to a minimum
- Handle photocell carefully (evacuated glass)

Analysis

From Einstein's photoelectric equation: $eV_0 = hf - \phi$

$$V_0 = (h/e)f - \phi/e$$

Plot **V_0 (y-axis) vs f (x-axis)**:

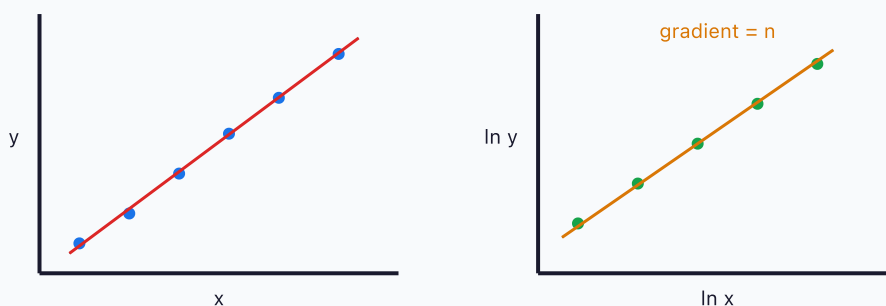
- Gradient = h/e (allows determination of Planck's constant h)
- y-intercept = $-\phi/e$ (allows determination of work function ϕ)
- x-intercept = threshold frequency $f_0 = \phi/h$

Source of Error	Type	Improvement
Stray light reaching photocell	Systematic	Darken the room; use shields around apparatus
Difficult to determine exact V_0 (current drops gradually)	Random	Plot I vs V and extrapolate to $I = 0$; use sensitive ammeter
Filters not monochromatic	Systematic	Use narrow-band filters or LEDs with known peak wavelength
Thermionic emission from cathode	Systematic	Keep cathode cool; subtract dark current

Exam Tip: The gradient h/e is the same for all metals — only the y-intercept (and hence work function and threshold frequency) changes. Below the threshold frequency, no photoelectrons are emitted regardless of intensity. This is a key piece of evidence for the particle nature of light.

Data Analysis Skills (used in every PAG)

Data Analysis Skills



Overview

This section covers data analysis techniques rather than a specific experiment. These skills are used in every PAG to process data, test relationships and determine unknown quantities.

Key Techniques

- **Linearising relationships:** If $y = kx^n$, plot $\ln y$ vs $\ln x$. Gradient = n , intercept = $\ln k$.
- **Log-log graphs:** Used when the power relationship is unknown.
- **Semi-log (ln-linear) graphs:** If $y = Ae^{kx}$, plot $\ln y$ vs x . Gradient = k .
- **Calculating uncertainties:** From gradients and intercepts of best-fit and worst-fit lines.
- **Error bars:** Plot on graphs using absolute uncertainties.

Method

1. Collect data from an experiment (often a student-designed investigation).

2. Process the data: calculate derived quantities, convert units.
3. Choose appropriate variables for plotting (to linearise the graph).
4. Plot the graph with error bars.
5. Draw best-fit line and worst-acceptable lines.
6. Determine gradient and intercept with uncertainties.
7. Use the gradient/intercept to determine the physical quantity of interest.

Safety

- Standard laboratory safety for whatever experiment is used
- Risk assessment specific to the chosen investigation

Analysis

Key formulae for linearisation:

Relationship	Plot	Gradient	Intercept
$y = mx + c$	y vs x	m	c
$y = kx^n$	$\ln y$ vs $\ln x$	n	$\ln k$
$y = Ae^{kx}$	$\ln y$ vs x	k	$\ln A$
$y = k/x$	y vs $1/x$	k	0

Source of Error	Type	Improvement
Anomalous data points	Random	Repeat measurements; identify and exclude outliers
Insufficient data range	Systematic	Collect data across the widest possible range
Poor graph plotting	Systematic	Use appropriate scales; fill at least half the graph paper

Uncertainty in gradient	Random	Use best-fit and worst-fit lines to determine uncertainty
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Exam Tip: The uncertainty in the gradient is found from: $\Delta\text{gradient} = (\text{gradient}_{\text{max}} - \text{gradient}_{\text{min}}) / 2$, where max and min come from the steepest and shallowest lines that pass through all error bars. Always include units in your gradient and intercept.